INCH-POUND
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DETAIL SPECIFICATION

VALVE, SOLENOID, THREE-WAY BYPASS (NAVAL SHIPBOARD USE)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three-way bypass valves, solenoid operated, for naval shipboard use as salinity-actuated dump valves.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Government specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for

MIL-DTL-1222 - Studs, Bolts, Screws and Nuts for Applications Where a High Degree of Reliability is Required; General Specification for

MIL-DTL-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4810

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MIL-PRF-8805	-	Switches and Switch Assemblies, Sensitive, Snap Action (Basic, Limit, Push Button and Toggle Switches), General Specification for
MIL-PRF-8805/1	-	Switches, Sensitive, SPDT, Unsealed
MIL-PRF-19500	-	Semiconductor Devices, General Specification for
MIL-PRF-19500/240	-	Semiconductor Device, Diode, Silicon, Rectifier, Types 1N645-1, 1N647-1, 1N649-1, 1N645UR-1, 1N647UR-1, 1N649UR-1, JAN, JANTX, and JANTXV
MIL-PRF-19500/427	-	Semiconductor Device, Diode, Silicon, Power Rectifier, Types 1N5614, 1N5616, 1N5618, 1N5620, 1N5622, 1N5614US, 1N5616US, 1N5618US, 1N5620US, 1N5622US, JAN, JANTX, JANTXV, JANS, JANHC, and JANKC
MIL-PRF-20042	-	Flanges, Pipe, Bronze (Silver Brazing)
MIL-DTL-24643	-	Cables, Electric, Low Smoke Halogen-Free, for Shipboard Use, General Specification for
MIL-DTL-24643/15	-	Cable, Electrical, -20 °C to +105 °C, 1000 Volts, Type LSDSGU

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Valves

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library at https://nll.navsup.navy.mil/default.cfm. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A125	-	Steel Springs, Helical, Heat Treated
ASTM A313/A313M	-	Wire Stainless Steel Spring
ASTM B61	-	Castings, Steam or Valve Bronze
ASTM B62	-	Metal Castings, Composition Bronze or Ounce
ASTM B127	-	Nickel-Copper Alloy (UNS N04400) Plate, Sheet, and Strip

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ASTM B164 - Rod, Bar, and Wire, Nickel-Copper Alloy

ASTM D2000 - Standard Classification System for Rubber Products in Automotive Applications

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AS40401 - Solenoid, Electrical, General Specification for

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.

3.2 Materials. Materials used in the construction of the valve shall be as shown in table I. Where no material is specified for a particular application, the material used shall be of the quality best suited for the purpose intended.

TABLE I. Materials.

Part	Material	Applicable document
Body	Valve bronze	ASTM B61
Disc	Nickel-copper-alloy	ASTM B127; QQ-N-281
Seat rings	Nickel-copper-alloy	ASTM B127; QQ-N-281
Stem	Nickel-copper-alloy	ASTM B164; QQ-N-281
Bonnet	Valve bronze	ASTM B61; ASTM B62
Yoke	Valve bronze, steel, or malleable iron	ASTM B61
Lever	Valve bronze	ASTM B61
Bolts and nuts	Nickel alloy	MIL-DTL-1222, Grade 400 or 405
Springs	Spring steel or electroplated zinc coated	ASTM A125; ASTM A313/A313M
Solenoid and switches	See 3.3.2 and 3.3.3	---

3.3 Construction. The valve shall be a three-way valve, arranged to allow flow from the outlet when the solenoid is energized via a relay from the salinity indicating system, and to be shifted by spring action into the dump position when the solenoid is de-energized. The valve shall be provided with a switch mechanically operated by the linkage connected to the valve stem, which will open the solenoid circuit upon solenoid de-energization, and which can only be closed when the valve is manually shifted to the outlet position. The valve shall be so constructed that it must be repositioned manually to allow flow to the outlet and that it may be tripped manually to the dump position. An additional switch shall be connected to the linkage in such a way that it will close a circuit to a signal light to indicate when the valve is in the dump position. The signal light shall not be a part of the valve. The valve shall be constructed for 50 pounds per square inch (psi) working pressure (see 4.2).

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3.3.1 Valve.

3.3.1.1 Ports. The inlet, outlet, and the dump port shall be flanged in accordance with MIL-PRF-20042, 150-pound class.

3.3.1.2 Bonnet. The valve bonnet shall be bolted to the body.

3.3.1.3 Disc. The valve disc shall be guided and positively secured to the valve stem, but in such a manner as to permit replacement.

3.3.1.4 Seats. The valve seats shall be renewable. The valve disc or sheet shall be provided with a resilient insert for seating. The resilient material shall be acrylonitrile in accordance with ASTM D2000 (BF 614 or BF 618). The term renewable, as applied herein, shall mean that valve parts known as seats shall be affixed or contained by a threaded element and be readily removed for replacement.

3.3.1.5 Stuffing box. A stuffing box shall be provided, packed with acceptable packing. The stuffing box shall be provided with glands, and shall be so constructed as to prevent excessive binding on stem. Glands of valves 2 inches and below may have gland nuts; those 2½ inches and above shall be of the bolted type.

3.3.2 Solenoid. Unless otherwise specified (see 6.2), solenoids shall be of the continuous duty type designated in accordance with SAE AS40401 for either alternating current (AC) operation or direct current (DC) operation through rectifier. Coils shall be class H. The electrical requirements for the solenoid shall be as follows:

a. Supply voltage range: 115 volts AC (VAC) ±20 percent (92 VAC to 138 VAC).

b. Ambient air temperature: 40 to 125 °F.

c. Frequency range: 60 hertz ±5.5percent (56.7 hertz to 63.3 hertz).

3.3.2.1 Rectifiers. A rectifier shall be provided for each solenoid designed for DC operation. The rectifier shall be built into the solenoid package and shall be silicon in accordance with MIL-PRF-19500 and MIL-PRF-19500/240 or MIL-PRF-19500/427. Rectifier circuits shall be of the full wave bridge type and designed for continuously energized operation from a nominal 115-VAC, 60-hertz power single-phase supply. The device shall operate satisfactorily within the voltage and frequency ranges specified in 3.3.2 and shall be removable for replacement.

3.3.2.2 Life and insulation. Solenoids shall be designed in accordance with the life requirements of SAE AS40401 for continuous energized operation of 168 hours at an ambient temperature of 125 °F and a supply voltage of 138 VAC maximum.

3.3.3 Switches. Switches shall be in accordance with MIL-PRF-8805 and MIL-PRF-8805/1, and shall be mounted in a watertight enclosure in accordance with MIL-DTL-2036, arranged for entry of cable conforming to type LSDSGU of MIL-DTL-24643 and MIL-DTL-24643/15.

3.3.4 Power. Electric equipment shall operate from a normal 115-volt, 60-hertz, single-phase power source.

3.3.5 Threaded parts. Threaded parts shall conform to FED-STD-H28. The construction shall be such that standard wrenches can be used throughout.

3.3.6 Welding and allied processes. Welding and allied processes shall be in accordance with S9074-AR-GIB-010/278.

3.3.7 Shock. Valves shall pass the requirements of high-impact shock tests in accordance with MIL-DTL-901.

3.3.8 Vibration. Valves shall pass the vibration requirements at the working pressure (see 3.3) in accordance with MIL-STD-167-1.

3.3.9 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.4 Valve size. Valve sizes shall be as specified (see 6.2).

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed when a first article sample is required (see 3.1). This inspection shall include the examination of 4.2.1, and the tests of 4.2.2 through 4.2.5.

TABLE II. First article inspection.

Inspection	Requirement	Test method	First article	Conformance
Visual and dimensional examination	3.2, 3.3, 3.3.1, 3.3.2, 3.3.3, 3.3.4, 3.3.5	4.2.1	X	X
Operation	3.3, 3.3.4	4.2.2	X	--
Shock and vibration	3.3.7, 3.3.8	4.2.3	X	--
Hydrostatic	3.3	4.2.4	X	X
Seat tightness	3.3.1.4	4.2.5	X	X

4.2.1 Examination. The various materials and parts of the components of each of the complete bypass valves shall be subjected to an examination to determine conformance with the various material specifications and with the requirements of this specification.

4.2.2 Operation. The valve shall be installed in a line supplied with fresh water with total dissolved solids below 500 parts per million (ppm) at the working pressure (see 3.3) from 28 to 103 °F, and the solenoid shall be connected to a source of the specified electrical current and voltage (see 3.3.4). The valve shall be subjected to 6,000 cycles of operation. A test for seat tightness shall be conducted at intervals of 1,000 cycles and at the completion of the cyclic test (a total of six seat tightness tests) in accordance with 4.2.5. The tripping mechanism and the electrical apparatus shall function as specified for the duration of the test.

4.2.3 Shock and vibration. The valve shall be shock tested at the working pressure (see 3.3) at full open and shock tested at the working pressure (see 3.3) on the inlet side at full closed. Equipment and test classification shall be grade A, hull mounted, principal unit, complete assembly, class I, type A. The valve shall be vibration tested at the working pressure (see 3.3) in accordance with MIL-STD-167-1. Upon completion of shock and vibration tests, the valves shall be subjected to the tests specified in 4.2.4 and 4.2.5.

4.2.4 Hydrostatic. The valve shall be subjected to a hydrostatic pressure of 100 psi gauge using fresh water with total dissolved solids below 500 ppm for strength and porosity. There shall be no visible leakage through any part of the valve body, the packing gland, or seats with the valve in either of its operating positions.

4.2.5 Leak test for seat tightness. Each valve shall be subjected to the leak tests using fresh water with total dissolved solids below 500 ppm as specified in 4.2.5.1 through 4.2.5.3. Test duration shall be 5 minutes at each test condition. The valve assembly shall be vented prior to each test to ensure that no air is present. No leakage shall be observed with the valve seat visible during the duration of each test.

4.2.5.1 Latched position, inlet port pressurized. The valve shall be tested in the latched position, inlet port pressurized to 100 psi and the dump port at 0 psi.

4.2.5.2 Dump position. The valve shall be tested in the dump position, inlet port pressurized to 100 psi and the outlet port at 0 psi.

4.2.5.3 Latched position, dump port pressurized. The valve shall be tested in the latched position, dump port pressurized to 100 psi and the inlet port at 0 psi.

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4.3 Conformance inspection. Conformance inspection shall be completed for every valve, and shall include the examination of 4.2.1, the hydrostatic test of 4.2.4, and the leak test for seat tightness of 4.2.5.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The valves covered by this specification are intended for use on reverse-osmosis units aboard U.S. Navy ships, requiring high reliability and the ability to operate in a marine environment. Further use of material procured to prior revisions of this specification should be discontinued and all future acquisitions should be to the current revision of this document.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article is required (see 3.1).

c. Solenoid type (see 3.3.2).

d. Valve size (see 3.4).

e. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Dump port

Salinity indicating system

Seat tightness

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4810-2017-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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